

have to be numbers; with the basics you learned in the first part of this book, you can analyze non-numerical data as well.

People use Python for data-intensive work in genetics, climate research, political and economic analysis, and much more. Data scientists have written an impressive array of visualization and analysis tools in Python, many of which are available to you as well. One of the most popular tools is Matplotlib, a mathematical plotting library. In this chapter, we'll use Matplotlib to make simple plots, such as line graphs and scatter plots. Then we'll create a more interesting dataset based on the concept of a random walk—a visualization generated from a series of random decisions.

We'll also use a package called Plotly, which creates visualizations that work well on digital devices, to analyze the results of rolling dice. Plotly generates visualizations that automatically resize to fit a variety of display devices. These visualizations can also include a number of interactive features, such as emphasizing particular aspects of the dataset when users hover over different parts of the visualization. Learning to use Matplotlib and Plotly will help you get started visualizing the kinds of data you're most interested in.

Installing Matplotlib

To use Matplotlib for your initial set of visualizations, you'll need to install it using `pip`, just like we did with `pytest` in Chapter 11 (see “Installing `pytest` with `pip`” on page 210).

To install Matplotlib, enter the following command at a terminal prompt:

```
$ python -m pip install --user matplotlib
```

If you use a command other than `python` to run programs or start a terminal session, such as `python3`, your command will look like this:

```
$ python3 -m pip install --user matplotlib
```

To see the kinds of visualizations you can make with Matplotlib, visit the Matplotlib home page at <https://matplotlib.org> and click **Plot types**. When you click a visualization in the gallery, you'll see the code used to generate the plot.

Plotting a Simple Line Graph

Let's plot a simple line graph using Matplotlib and then customize it to create a more informative data visualization. We'll use the square number sequence 1, 4, 9, 16, and 25 as the data for the graph.

To make a simple line graph, specify the numbers you want to work with and let Matplotlib do the rest:

```
mpl_squares.py import matplotlib.pyplot as plt  
  
squares = [1, 4, 9, 16, 25]
```
